

📌 Part 1: What is Correlation?

◆ Definition

👉 Correlation means relationship between two variables.

👉 It shows:

- How one value changes with another
 - Whether variables are related
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📌 Part 2: Types of Correlation

Type	Meaning
Positive Correlation	Both increase together
Negative Correlation	One increases, other decreases
No Correlation	No relationship

📌 Part 3: Real-Life Examples

◆ Positive Correlation

👉 Study hours $\uparrow$ $\rightarrow$ Marks $\uparrow$

◆ Negative Correlation

👉 Exercise $\uparrow$ $\rightarrow$ Weight $\downarrow$

◆ No Correlation

👉 Shoe size $\leftrightarrow$ Exam marks

📌 Part 4: Theory Answers (IMPORTANT)

◆ 1. What is Correlation?

👉 Correlation measures relationship between variables.

◆ 2. What is Positive Correlation?

👉 When both variables increase together.

◆ 3. What is Negative Correlation?

👉 When one variable increases and the other decreases.

◆ 4. What is Heatmap?

👉 Heatmap is a colored graphical representation of data.

👉 Used to visualize correlation values.

◆ 5. What is Pairplot?

👉 Pairplot shows relationships between multiple variables together.

📌 Part 5: Import Libraries

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

📌 Part 6: Create Dataset

```
data = {
    "StudyHours": [1, 2, 3, 4, 5],
    "Marks": [40, 50, 60, 70, 80],
    "SleepHours": [8, 7, 6, 5, 4]
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

📌 Part 7: Finding Correlation

```
print(df.corr())
```

👉 Output values range from:

- +1 → strong positive correlation
 - -1 → strong negative correlation
 - 0 → no correlation
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📌 Part 8: Heatmap

◆ Visualize Correlation

```
sns.heatmap(df.corr(), annot=True)
```

```
plt.show()
```

◆ Understanding Parameters

Parameter Meaning

corr() Finds correlation

annot=True Shows values inside boxes

📌 Part 9: Pairplot

```
sns.pairplot(df)
```

```
plt.show()
```

👉 Pairplot automatically creates:

- Scatter plots
 - Distribution plots
 - Relationship analysis
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📌 Part 10: Scatterplot for Correlation

```
sns.scatterplot(x="StudyHours", y="Marks", data=df)
```

```
plt.show()
```

👉 Shows positive relationship visually.

📌 Part 11: Correlation Matrix Visualization

	StudyHours	Marks	SleepHours
StudyHours	1.0	0.95	-0.90
Marks	0.95	1.0	-0.85
SleepHours	-0.90	-0.85	1.0

📌 Part 12: Real-World Thinking

👉 In AI/Data Science:

- Detect feature relationships
- Remove unnecessary features
- Analyze patterns in dataset

👉 Used in:

- Predictive analysis
- Healthcare datasets
- Sales forecasting
- Student performance analysis

Part 13: Why Correlation is Important in ML?

 Machine Learning models learn patterns from relationships.

If features are strongly related:

- Prediction becomes easier
- Analysis improves

Part 14: Heatmap Color Understanding

Color	Meaning
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Dark color	Strong correlation
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Light color	Weak correlation
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Tasks for Day 33

Task 1: Theory (Write Answers)

1. What is correlation?
2. Difference between positive and negative correlation
3. What is heatmap?
4. What is pairplot?
5. Why correlation is important in ML?

Task 2: Correlation Practice

1. Create dataset
2. Find correlation matrix
3. Analyze relationships

Task 3: Heatmap Practice

1. Create heatmap
 2. Show annotations
 3. Analyze strong correlations
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Task 4: Pairplot Practice

1. Create pairplot
 2. Analyze distributions
 3. Compare variables
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Task 5: Scatter Plot Practice

1. Study hours vs marks
 2. Salary vs experience
 3. Temperature vs sales
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Task 6: Dataset Analysis

1. Student dataset
2. Employee dataset
3. Hospital dataset